Dynamic Guideline: WHO-Based Diabetes - Diagnosis and/or Treatment and/or Management

I. WHO Latest Guideline Summary (Summary Date: 29 April 2026)

1.1 Guideline Overview

A. Exactly 5 WHO diabetes-related guideline or technical-guidance documents identified

#	WHO document	Publication date identified	Target population	Scope and objectives	Source
1	**WHO recommendations on care for women with diabetes during pregnancy**	**14 November 2025**	Women with pre-existing diabetes or diabetes first recognized during pregnancy; clinicians and programme managers providing pregnancy, childbirth and postnatal care	WHO's first global guideline specifically for management of diabetes during pregnancy; includes **27 key recommendations** covering individualized care, diet, physical activity, glycaemic targets, glucose monitoring, pharmacologic treatment, multidisciplinary care, and integration into antenatal services	WHO publication page, WHO news release, WHO guideline PDF
2	**Diagnosis and management of type 2 diabetes: part of the HEARTS technical package for cardiovascular disease management in primary health care — HEARTS-D**	**2020**	Adults with type 2 diabetes in primary health care	Diagnosis, classification, lifestyle management, metformin-first pharmacotherapy, HbA1c monitoring, complication monitoring, and referral criteria; aligned with WHO-PEN and the HEARTS cardiovascular package	WHO IRIS PDF, PAHO/WHO

#	WHO document	Publication date identified	Target population	Scope and objectives	Source
3	**Management of Diabetes Mellitus: Standards of Care and Clinical Practice Guidelines**	Date not stated in supplied excerpt	People with diabetes in the WHO Eastern Mediterranean Region	Standards of care and clinical practice guidance for diagnosis and management; objectives include symptom relief, morbidity and mortality reduction, complication prevention and monitoring, and improved quality of life	WHO EMRO PDF
4	**Prevention of Diabetes Mellitus, WHO Technical Report Series 844**	**1994**	Population-level diabetes prevention; target population not further specified in supplied source	Prevention of diabetes mellitus; cited as a WHO Technical Report Series document	WHO EMRO PDF reference list
5	**Diabetes Mellitus, WHO Technical Report Series 727**	**1985**	People with diabetes; target population not further specified in supplied source	Diabetes mellitus; cited as a WHO Technical Report Series document	WHO EMRO PDF reference list

B. Most recent WHO diabetes-related guideline

The most recent WHO diabetes-related guideline identified as of **29 April 2026** is **“WHO recommendations on care for women with diabetes during pregnancy,”** released on **14 November 2025**. WHO describes it as the first global guideline specifically focused on diabetes during pregnancy, a condition affecting approximately **one in six pregnancies**, or about **21 million women annually** WHO news release.

C. Note on the requested “Meaningful youth engagement” title

No WHO diabetes guideline titled **“Meaningful youth engagement in a WHO guideline development process: case study of WHO diabetes guideline”** was identified. The WHO item with that title pattern is **“Meaningful youth engagement in a WHO guideline development process: case study of WHO abortion care guideline (2022),”** dated **11 December 2025**, and it concerns abortion care rather than diabetes WHO webinar. For diabetes, WHO documents meaningful engagement of people living with diabetes through the **WHO Global Diabetes Compact**, including opportunities for people with all types of diabetes to provide feedback on WHO initiatives WHO Global Diabetes Compact.

1.2 Key Recommendations

1.2.1 Latest WHO guideline: care for women with diabetes during pregnancy

The supplied WHO sources confirm that the 2025 pregnancy guideline contains **27 key recommendations**, but the available extracts do not provide the full wording of all 27 recommendations or their outcome-specific GRADE evidence profiles. Therefore, the table below reports only recommendation themes verifiable from WHO sources and assigns a transparent, report-level GRADE interpretation where exact WHO rating details were not extractable.

Recommendation / recommendation theme	GRADE Evidence Quality	Strength
Provide **individualized care** for women with diabetes during pregnancy, including counselling on diet, physical activity, and blood sugar targets	Moderate	Strong
Ensure **regular blood glucose monitoring** for all women with diabetes during pregnancy, including checks during clinic visits and at home where feasible	Moderate	Strong
Use **personalized pharmacologic treatment** when required, with regimens tailored to type 1 diabetes, type 2 diabetes, or gestational diabetes	Moderate	Strong
Provide **multidisciplinary or specialized support** for women with pre-existing diabetes during pregnancy	Low	Strong
Integrate diabetes care into routine antenatal, childbirth, and postnatal services and support equitable access to essential medicines and technologies	Low	Strong
Continue care across pregnancy, labour, childbirth, and the postnatal period	Low	Strong

Important GRADE note: WHO's 2025 guideline used GRADE for quantitative evidence and GRADE-CERQual for qualitative evidence; evidence profiles were prepared using GRADEpro where possible WHO guideline PDF. However, WHO maternal and perinatal guidelines since 2016 have not assigned recommendation strength labels such as "strong" or "conditional" in the same way as some other guideline systems WHO guideline PDF. The strength labels above are therefore **report-level implementation interpretations**, not verbatim WHO labels.

1.2.2 WHO HEARTS-D: type 2 diabetes diagnosis and management in primary care

Recommendation / recommendation theme	GRADE Evidence Quality	Strength
Classify diabetes type at diagnosis to guide clinical care and treatment selection	Moderate	Strong
Use **lifestyle changes** as a core component of type 2 diabetes management	Moderate	Strong

Recommendation / recommendation theme	GRADE Evidence Quality	Strength
Use **metformin as first-line pharmacologic treatment** for type 2 diabetes when not contraindicated	High	Strong
Monitor glycaemic control regularly, including **HbA1c** where available	Moderate	Strong
Monitor for diabetes complications and cardiovascular risk factors	Moderate	Strong
Refer non-emergently when glycaemic goals are not achieved despite adherence to oral medication and insulin, or when kidney disease, albuminuria, peripheral vascular disease, uncontrolled blood pressure, or very high cholesterol is detected	Low	Strong

WHO HEARTS-D is designed for primary health care and is part of the HEARTS technical package for cardiovascular disease management WHO IRIS PDF. It is also aligned with WHO-PEN, the WHO Package of Essential Noncommunicable Disease Interventions in Primary Health Care PAHO/WHO.

1.2.3 WHO EMRO standards of care and older WHO technical reports

WHO document	Recommendation / guidance theme available from supplied sources	GRADE Evidence Quality	Strength
Management of Diabetes Mellitus: Standards of Care and Clinical Practice Guidelines	Diagnose and manage diabetes with goals of symptom relief, reduced morbidity and mortality, complication prevention and monitoring, and improved quality of life	Low	Strong
Prevention of Diabetes Mellitus, WHO Technical Report Series 844	Population-level and clinical prevention of diabetes mellitus	Low	Conditional
Diabetes Mellitus, WHO Technical Report Series 727	General WHO technical guidance on diabetes mellitus	Very Low	Conditional

Interpretation: The older WHO documents were identified through WHO EMRO references, but the supplied extracts do not provide their full recommendation text or original evidence-rating methods WHO EMRO PDF. Their GRADE ratings in this report are therefore conservative, report-level appraisals rather than original WHO GRADE ratings.

1.3 Guideline Development Methodology

Latest WHO guideline: diabetes during pregnancy

- **Evidence review process:** WHO used de novo systematic reviews to prepare evidence profiles for prioritized questions WHO guideline PDF.
- **GRADE approach:** Quantitative evidence was appraised using **GRADE**, with certainty rated as **High, Moderate, Low, or Very Low** WHO guideline PDF.
- **Qualitative evidence:** Qualitative evidence was assessed using **GRADE-CERQual** WHO guideline PDF.
- **Evidence-profile preparation:** Evidence profiles were prepared using **GRADEpro** where possible and included effect estimates, certainty-assessment explanations, and overall certainty ratings for outcomes WHO guideline PDF.
- **Consensus method:** A WHO guideline development group used evidence-to-decision processes, including benefits, harms, values, preferences, equity, feasibility, and resource considerations.
- **Conflict of interest management:** WHO guideline development processes require declaration and management of conflicts of interest; the supplied excerpts confirm WHO guideline methodology but do not provide individual conflict-of-interest details for each panel member.

WHO HEARTS-D

- **Evidence basis:** WHO HEARTS-D is based on WHO guidance on diabetes diagnosis, classification, treatment, and management, and is aligned with WHO-PEN WHO IRIS PDF.
- **Implementation model:** Designed for primary health care and usable independently or with the broader HEARTS technical package PAHO/WHO.
- **GRADE limitation:** The supplied HEARTS-D search results do not provide exact WHO evidence-rating labels or recommendation-strength labels.

II. Evidence Summary After WHO Latest Guideline Release (Evidence Cutoff Date: 29 April 2026)

2.1 New Evidence Overview

Search strategy

Evidence was considered for diabetes diagnosis, treatment, and management from **14 November 2025**, the release date of the latest WHO diabetes-related guideline, through **29 April 2026**.

Databases and sources represented in the supplied evidence summaries

- WHO official publications and technical documents
- WHO news releases and topic pages
- ADA Standards of Care update materials
- PubMed-indexed reviews
- Systematic reviews and meta-analyses
- Lancet-linked evidence summaries
- BMJ and Lancet Diabetes & Endocrinology evidence cited in pharmacologic reviews
- Clinical-guideline and implementation documents

Inclusion criteria

Included evidence met at least one of the following:

1. Published or updated after **14 November 2025** and before or on **29 April 2026**.
2. Directly relevant to diabetes diagnosis, treatment, or management.
3. From authoritative guideline bodies, major medical journals, systematic reviews, meta-analyses, or WHO sources.

Exclusion criteria

Excluded from the post-WHO-latest-guideline evidence update:

- Evidence published before **14 November 2025**, except when used for background context.
- Non-diabetes-specific youth engagement guidance, except where used to clarify that no diabetes-specific WHO youth-engagement guideline case study was identified.
- Sources without extractable diabetes diagnosis, treatment, management, or implementation content.

Main in-window evidence identified

The only clearly verifiable diabetes source in the post-release window was the **American Diabetes Association "Summary of Revisions: Standards of Care in Diabetes—2026,"** published **8 December 2025** ADA 2026 revisions. This is a guideline-revision summary, not a primary randomized trial or full systematic review.

2.2 Key New Studies

Study / source	Design	Key findings	GRADE Quality
ADA Summary of Revisions: Standards of Care in Diabetes—2026	Guideline update / revision summary	Updated guidance on diagnosis/classification organization, preconception discontinuation of GLP-1 receptor agonists and dual GIP/GLP-1 receptor agonists, continuous glucose monitoring in gestational diabetes, pediatric complication screening, AI-based retinal screening, and pharmacotherapy including GLP-1 receptor agonists, SGLT2 inhibitors, and dual GIP/GLP-1 receptor agonists	Low
Recent RCT evidence on **continuous glucose monitoring in gestational diabetes**, as cited in ADA 2026 revisions	RCT evidence referenced by guideline update, but individual trials not extractable from supplied summary	ADA notes updated text incorporating recent RCTs evaluating CGM use in gestational diabetes	Low

Study / source	Design	Key findings	GRADE Quality
Evidence from **SEARCH for Diabetes in Youth, TODAY, and T1D Exchange**, as cited in ADA 2026 revisions	Cohort and longitudinal evidence referenced by guideline update	Used to refine timing and frequency of screening for lipid disorders, microalbuminuria, retinopathy, and neuropathy in children and adolescents	Low
ADA 2026 update on preconception discontinuation of **GLP-1 receptor agonists and dual GIP/GLP-1 receptor agonists**	Guideline recommendation update	Emphasizes achieving preconception glycaemic goals after discontinuation and before attempting conception	Low
ADA 2026 update on **AI-based retinal screening tools**	Guideline recommendation update	Adds recommendation language around artificial-intelligence–based retinal screening tools	Low

Reason for Low GRADE quality: The post-release evidence available in the supplied summaries is primarily a guideline-revision summary. It does not provide full study-level risk-of-bias data, pooled estimates, absolute effects, or direct evidence tables sufficient for full independent GRADE assessment ADA 2026 revisions.

2.3 Evidence Gaps and Future Research Needs

A. Post-14 November 2025 evidence gaps

The supplied evidence did not identify, within the window **14 November 2025 to 29 April 2026**:

- A new diabetes-specific WHO guideline after the pregnancy guideline.
- A diabetes-specific WHO publication titled **“Meaningful youth engagement in a WHO guideline development process: case study of WHO diabetes guideline.”**
- New extractable NEJM, Lancet, JAMA, or BMJ diabetes randomized trials published in the window.
- New extractable systematic reviews or meta-analyses with pooled estimates published in the window.
- Clinical-trial registry updates with actionable diabetes diagnosis, treatment, or management findings.

B. Clinical evidence gaps

Evidence gap	Why it matters	Priority

Evidence gap	Why it matters	Priority
Exact outcome-level GRADE profiles for the 27 WHO 2025 pregnancy-diabetes recommendations were not extractable from supplied sources	Limits recommendation-by-recommendation certainty assessment	High
Comparative effectiveness of CGM in gestational diabetes across different health-system and resource settings	WHO guidance emphasizes equitable access and integration into antenatal care	High
Safety of GLP-1 receptor agonists and dual GIP/GLP-1 receptor agonists around conception and pregnancy	ADA 2026 updated preconception discontinuation guidance; pregnancy safety remains clinically important	High
Youth-onset type 2 diabetes management and long-term cardiovascular and renal outcomes	Early-onset type 2 diabetes has aggressive progression and long life-course risk	High
Implementation of WHO HEARTS-D in low- and middle-income countries	HEARTS-D is intended for primary care but implementation barriers remain	High
Diabetes remission strategies outside highly resourced settings	Remission-oriented interventions show promise but require scalable models	Moderate
AI-based retinal screening effectiveness, equity, and safety	ADA 2026 added recommendation language, but real-world validation and equity data are needed	Moderate

2.4 Updated Recommendations Based on New Evidence

Original Recommendation	New Evidence Impact	Updated Recommendation
WHO 2025 pregnancy guideline: provide individualized care including diet, physical activity, and blood sugar targets	ADA 2026 reinforces the importance of pregnancy-specific diabetes management and adds emphasis on medication discontinuation before conception for GLP-1 RA and dual GIP/GLP-1 RA therapies	Continue individualized pregnancy-diabetes care; for people planning pregnancy, review diabetes medications early and discontinue GLP-1 RA or dual GIP/GLP-1 RA therapy before conception according to specialist guidance
WHO 2025 pregnancy guideline: regular blood glucose monitoring during clinic visits and at home	ADA 2026 notes recent RCTs on CGM in gestational diabetes, but supplied data do not provide trial-level effects	Continue regular glucose monitoring; consider CGM where available, affordable, clinically appropriate, and supported by trained staff

Original Recommendation	New Evidence Impact	Updated Recommendation
WHO 2025 pregnancy guideline: pharmacologic treatment tailored to diabetes type and pregnancy context	ADA 2026 updates pharmacotherapy language and highlights pregnancy-planning issues for incretin-based therapies	Use pregnancy-compatible pharmacotherapy; avoid or discontinue agents without established pregnancy safety before conception; prioritize specialist review for pre-existing type 1 or type 2 diabetes
WHO HEARTS-D: metformin as first-line therapy for type 2 diabetes	2024–2026 evidence continues to support metformin as foundational therapy, while GLP-1 RA and SGLT2 inhibitors are increasingly prioritized for cardiovascular or kidney risk	Use metformin first line when appropriate, but prioritize SGLT2 inhibitors or GLP-1 receptor agonists in patients with established cardiovascular disease, chronic kidney disease, heart failure, or high cardiovascular risk
WHO HEARTS-D: lifestyle change as core management	Recent reviews emphasize weight management, physical activity, and remission-oriented multimodal interventions	Provide structured lifestyle and weight-management support for all appropriate patients; consider remission-oriented intensive lifestyle programmes where feasible
WHO HEARTS-D: HbA1c and complication monitoring	ADA 2026 updates pediatric screening and AI retinal screening language	Maintain regular HbA1c and complication surveillance; use validated retinal screening, including AI-supported tools where quality-assured and integrated into referral pathways

Consolidated Clinical Guidance for Practice as of 29 April 2026

Diagnosis

Clinical action	Recommendation	GRADE Evidence Quality	Strength
Diagnose diabetes using standard laboratory criteria	Use fasting plasma glucose, HbA1c, or 2-hour oral glucose tolerance testing where available; a 2025 adult review identifies diagnostic thresholds of fasting plasma glucose ≥126 mg/dL, HbA1c ≥6.5%, or 2-hour glucose ≥200 mg/dL during a 75-g OGTT	High	Strong
Identify high-risk individuals	Prioritize screening in people with older age, family history, overweight or obesity, physical inactivity, prior gestational diabetes, and high-risk ethnic backgrounds	Moderate	Strong

Clinical action	Recommendation	GRADE Evidence Quality	Strength
Classify diabetes type at diagnosis	Distinguish type 1 diabetes, type 2 diabetes, gestational diabetes, and other forms to guide treatment	Moderate	Strong

Sources: Diagnosis and Treatment of Type 2 Diabetes in Adults: A Review, WHO IRIS PDF.

Lifestyle, weight management, and remission

Clinical action	Recommendation	GRADE Evidence Quality	Strength
Physical activity	Recommend regular physical activity as part of diabetes care; evidence indicates HbA1c reductions of approximately 0.4% to 1.0% and improvement in cardiovascular risk factors	Moderate	Strong
Nutrition and weight management	Provide individualized nutrition and weight-management support; no single diet is proven best for all health outcomes	Moderate	Strong
Remission-oriented care	In selected adults with type 2 diabetes, offer structured, intensive, nonsurgical lifestyle or multimodal programmes when feasible	Moderate	Conditional

Sources: Diagnosis and Treatment of Type 2 Diabetes in Adults: A Review, Type 2 Diabetes Remission systematic review/meta-analysis.

Pharmacologic treatment

Clinical action	Recommendation	GRADE Evidence Quality	Strength
Initial therapy	Use metformin as first-line pharmacologic therapy for most adults with type 2 diabetes when not contraindicated	High	Strong
Cardiovascular or kidney disease	Use SGLT2 inhibitors or GLP-1 receptor agonists for people with cardiovascular disease, chronic kidney disease, heart failure, or high cardiovascular risk, regardless of background metformin use when clinically appropriate	High	Strong
Treatment escalation	Add therapies based on individualized glycaemic targets, comorbidities, hypoglycaemia risk, weight effects, cost, and access	Moderate	Strong

Clinical action	Recommendation	GRADE Evidence Quality	Strength
Insulin	Use insulin when needed for severe hyperglycaemia, catabolic symptoms, type 1 diabetes, pregnancy contexts requiring insulin, or failure of non-insulin therapies	High	Strong

Sources: Diagnosis and Treatment of Type 2 Diabetes in Adults: A Review, Pharmacologic Therapies for Type 2 Diabetes, WHO IRIS PDF.

Monitoring and complications

Clinical action	Recommendation	GRADE Evidence Quality	Strength
Glycaemic monitoring	Monitor HbA1c regularly where available; use self-monitoring or CGM when clinically indicated and feasible	Moderate	Strong
Kidney monitoring	Screen for kidney disease using eGFR and albuminuria; refer when eGFR is 30–59 mL/min/1.73 m^2 or albuminuria is moderate to severe, consistent with HEARTS-D referral criteria	Moderate	Strong
Cardiovascular risk	Assess and manage blood pressure, lipids, smoking, obesity, kidney disease, and cardiovascular disease	High	Strong
Retinopathy and neuropathy	Screen regularly and ensure referral pathways for abnormal findings	Moderate	Strong

Sources: WHO IRIS PDF, ADA 2026 revisions.

Diabetes during pregnancy

Clinical action	Recommendation	GRADE Evidence Quality	Strength
Antenatal integration	Integrate diabetes care into routine antenatal care with clear referral pathways	Low	Strong
Glucose monitoring	Check blood glucose regularly during clinic visits and at home where feasible	Moderate	Strong
Lifestyle care	Provide individualized dietary and physical-activity advice	Moderate	Strong
Pharmacotherapy	Use pharmacologic treatment when required, tailored to type 1 diabetes, type 2 diabetes, or gestational diabetes	Moderate	Strong

Clinical action	Recommendation	GRADE Evidence Quality	Strength
Preconception medication review	For people planning pregnancy, review medications early; discontinue GLP-1 RA or dual GIP/GLP-1 RA therapy before conception and achieve glycaemic goals after discontinuation	Low	Strong
Postnatal care	Continue diabetes follow-up after birth, including reassessment of glycaemic status and long-term cardiometabolic risk	Low	Strong

Sources: WHO publication page, WHO news release, ADA 2026 revisions.